

Konstantinos Oxouzoglou

Junior Robotics, Embedded Systems, Software & AI Engineer

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PROFILE

Computer Science graduate with all degree requirements completed in July 2026, interested in building intelligent systems across embedded hardware, real-time software, machine learning and cross-platform applications. Strong project-based experience turning technical requirements into tested prototypes and maintainable software.

SELECTED PROJECTS

BIOS Beep Code Analyzer

BSC THESIS • EMBEDDED SYSTEMS

- Designed an Arduino Uno diagnostic device that decodes motherboard beep signals and presents hardware fault guidance on a 128×64 OLED.
- Implemented approximately 1,900 lines of C++ firmware using edge detection, dual finite-state machines, EEPROM persistence and BIOS-specific timing.
- Supported AMI, Award and Phoenix patterns; passed 7/7 defined hardware test scenarios.

Heart Disease Neural Network

ACADEMIC AI PROJECT

- Built a TensorFlow/Keras MLP for binary classification of the Cleveland Heart Disease dataset from 13 clinical features.
- Achieved 95.56% accuracy on a 45-sample test split, with precision, recall and F1 score around 0.95–0.96.
- Applied data cleaning, standardisation, stratified splitting, dropout, batch normalisation and early stopping.

FoodRush

FULL-STACK & MOBILE PROJECT

- Developed React web and Expo/React Native clients sharing a Node.js/Express REST API and LowDB data layer.
- Implemented authentication, cart and checkout flows, order tracking, geolocation, WebAuthn and mobile biometrics.

AI Keychain

EMBEDDED EDGE-AI CONCEPT • IN PLANNING

- Designed a modular ESP32-S3 architecture for wake word, speech, hybrid local/cloud AI, I2S audio and power management.

EDUCATION

BSc (Hons) Computer Science

University of East London

2022–2026 • COURSEWORK COMPLETED JULY 2026

Final-year thesis focused on embedded diagnostics and real-time BIOS acoustic signal analysis.

TECHNICAL SKILLS

Embedded

C/C++, Arduino, finite-state machines, I2C, I2S, signal processing

AI & Data